Thomas C. Fraser

Curriculum Vitae

June 17, 2026

Personal

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Address	Strynøgade 5, 1.08, 2100 Copenhagen, Denmark
Citizenship	Canadian
Date of Birth	27/11/1994

Positions & education

Jan. 2024 – present	Postdoctoral Researcher *Department of Mathematics, University of Copenhagen*
Sept. 2018 – Sept. 2023	Ph.D., Physics Thesis: An estimation theoretic approach to quantum realizability problems *Perimeter Institute For Theoretical Physics, Waterloo, ON*
Aug. 2017 – Jun. 2018	M.Sc., Theoretical Physics, Quantum Information Thesis: A Combinatorial Solution to Causal Compatibility *Perimeter Scholars International Program, Perimeter Institute For Theoretical Physics, Waterloo, ON*
Sept. 2012 – Apr. 2017	B.Sc., Mathematical Physics, Astrophysics Specialization *University of Waterloo, Waterloo, ON*

Papers

2026 Thomas C. Fraser, V. Blakaj, R. Rubboli, F. Huber, M. Fanizza
"Complete entanglement detection using polynomial invariants"
arXiv:2606.16712

2026 A. Botero, M. Christandl, Thomas C Fraser, I. Leigh, H. Nieuwboer
"On quantum functionals for higher-order tensors"
arXiv:2604.18283

2025 A. Alekseev, M. Christandl, Thomas C. Fraser
"The tetrahedral Horn problem and asymptotics of $U(n)$ $6j$ symbols"
arXiv:2510.04877

2023 Thomas. C. Fraser
"An estimation theoretic approach to quantum realizability problems"
Ph.D. Thesis. arXiv:2402.10902

2023 D. Schmid, Thomas. C. Fraser, R. Kunjwal, A. B. Sainz, E. Wolfe, R. W. Spekkens
"Understanding the interplay of entanglement and nonlocality: motivating and developing a new branch of entanglement theory"
Quantum 7, 1194

2022 Thomas. C. Fraser
"A sufficient family of necessary inequalities for the compatibility of quantum marginals"
arXiv preprint arXiv:2211.00685

2020 Thomas. C. Fraser
"A Combinatorial Solution to Causal Compatibility"
Journal of Causal Inference 8(1), 22-53

2018 J. R. Rinehart, J. H. Béjanin, Thomas. C. Fraser, M. Mariantoni
"Machine Learning Peeling and Loss Modelling of Time-Domain Reflectometry"
arXiv preprint arXiv:1804.04756

2018 Thomas. C. Fraser, E. Wolfe
"Causal Compatibility Inequalities Admitting of Quantum Violations in the Triangle Scenario"
Physical Review A, 98(2), 022113

Teaching

Block 3 2025,2026	Introduction to Representation Theory - University of Copenhagen *Lecturer*
Block 3 2024	Introduction to Representation Theory - University of Copenhagen *Teaching Assistant*
Winter 2021	Quantum Information - Perimeter Scholars International (PSI) Program *Teaching Assistant*
Winter 2021	Quantum-classical gaps in causal structures - PSI Winter School *Student project co-lead*
Fall 2018	Physics 1 - Phys 111 - University of Waterloo *Teaching Assistant*

Non-academic Experience

September 2015 – January 2016	*Research & Development Data Scientist* Sysomos, Toronto, ON, Canada
September 2014 – May 2015	*Video Game Developer* Lunarch Studios, Waterloo, ON, Canada
January 2014 – May 2014	*Mathematics Tutor* Humber College, Toronto, ON, Canada
June 2011 – August 2011	*Solar Panel Technician* OVG Solar, Inc, Renfew, ON, Canada

Programming

Languages	C, C++, Python, Mathematica, Lua, Matlab, Actionscript, JavaScript

Updated editions of this CV available at https://github.com/tcfraser/curriculum-vitae/raw/master/cv.pdf